

Homework Assignment

Submission and Evaluation Guidelines

Deadline

Submission deadline: 29 May 2026, 23:59.

This date is the last teaching day of the spring term at BME in the 2025/2026 academic year. Late submissions may not be accepted unless there is a previously agreed exceptional reason.

1 Who receives the assignment?

- Students who completed the midterm will receive their individualized homework assignment by email.
- Students who **did not write the midterm** but plan to take the make-up exam must **request** their homework assignment by email.
- Request and submission email address: vizermatebme@gmail.com.

2 Submission format

Preferred format

Please submit your solution preferably as a **Jupyter Notebook** file, that is, an `.ipynb` file, with all code cells executed and all outputs visible.

Other formats are also accepted if necessary, for example:

- an exported **PDF** version of the notebook;
- an **R Markdown** or **Quarto** document;
- a clearly structured script together with a written explanation;
- other statistical software output, for example **SPSS**, if the interpretation is written clearly.

Important: the email subject and every submitted filename must include your full name. If you submit several files, put them into one clearly named folder or `.zip` file.

3 Programming language and tools

- It is recommended to use R or Python.
- Other tools are accepted if they are appropriate for the statistical analysis.

- The most important requirement is not the software itself, but that the analysis is correct, understandable, and well explained.

4 What a good solution must contain

For each task, your solution should be written in a way that another student could read it and understand both **what you did** and **why you did it**.

1. **Problem statement.** Briefly state which problem you are solving and which data file you are using.
2. **Choice of method.** Explain which statistical method you use and why it is suitable for the problem.
3. **Hypotheses, if relevant.** Clearly formulate the null hypothesis and the alternative hypothesis.
4. **Assumptions.** State the assumptions of the method and check them when appropriate, for example normality, equality of variances, independence, or repeated-measures structure.
5. **Calculations and output.** Show the relevant code and output, but do not paste large, unexplained blocks of software output.
6. **Interpretation.** Explain the results in ordinary language and answer the original question.

5 Every numerical result must be explained

Main evaluation principle

Every important number in your solution must be explained. The goal is to show that you understand the statistical meaning of the result, not only that you can run the command.

The purpose of the homework is to practise statistical reasoning. A good solution is not merely a collection of commands and outputs: it is a clear explanation of the analysis and its meaning.

For example, do **not** only write:

```
p = 0.032, therefore significant.
```

Instead, explain what the number means, for example:

The p-value is 0.032. This means that, if the null hypothesis were true, observing a test statistic at least this extreme would have probability approximately 3.2%. Since this is below the usual 5% significance level, we reject the null hypothesis and conclude that the data provide evidence for a difference between the groups.

Typical numbers that must be interpreted include:

- sample means, medians, standard deviations, and sample sizes;
- test statistics, degrees of freedom, and p-values;

- confidence intervals and their practical meaning;
- effect sizes, if used;
- ANOVA tables, post-hoc comparisons, and model coefficients;
- diagnostic quantities and plots, for example residual plots or normality checks.

6 Figures, tables, and code output

- Every figure should have a short explanation: what is shown, what pattern is visible, and how it supports the analysis.
- Every table should be introduced and interpreted. Do not leave a table without explanation.
- Code should be readable and organized into small blocks.
- Avoid unnecessary output. Include only what helps answer the statistical question.

7 Defence of the solution

Oral defence

Each student must be able to **defend their own solution** during the exam by answering questions about the statistical analyses performed.

- The defence checks whether you understand the methods, assumptions, numerical results, and interpretations in your submitted work.
- Each student must achieve at least **40%** on the defence part.
- Students who submit earlier may defend their solution during the spring term, if this is arranged in advance.

8 Submission checklist

Before sending your work, check the following:

- ☐ My full name is included in the filename and in the email subject.
- ☐ The notebook or document contains all tasks assigned to me.
- ☐ All code cells have been executed and the outputs are visible.
- ☐ Every statistical method is named and justified.
- ☐ All important assumptions are stated and checked when appropriate.
- ☐ Every important numerical result is explained in words.
- ☐ The final conclusion answers the original question of each task.
- ☐ Figures and tables are readable and interpreted.